

Shakura & Sunyaev (1973), “Black Holes in Binary Systems: Observational Appearance”

In this review, I will go through the physical picture developed in the paper. The paper starts with explaining how the Blackholes are detected. As they do not radiate neither electromagnetic waves nor gravitational waves, they can only be found from the gravitational influence that they produce on a neighbouring star or the gas medium which surrounds it.

From a Binary System to an Accretion Disk

Up to 50% of stars are in binary systems. What will be the consequences of the existence of a black hole in a binary system if matter flows strongly outwards from the visible star? In this paper, two types of matter capture are explained for a binary system consisting of a companion star and a black hole.

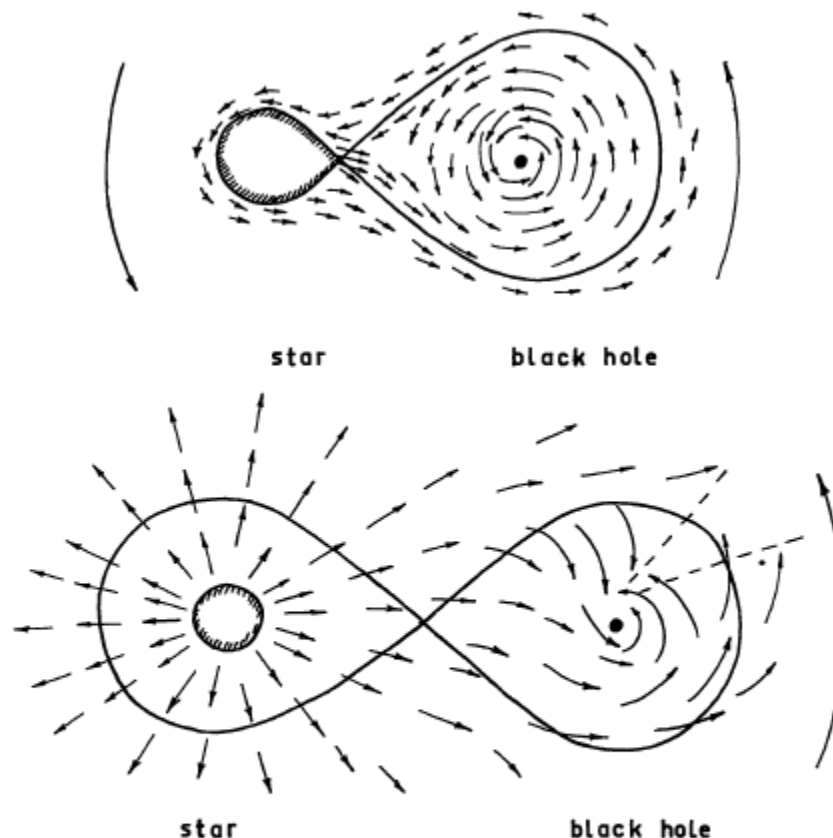


Fig. 1. Two regimes of matter capture by a collapsar: a) a normal companion fills up its Roche lobe, and the outflow goes, in the main, through the inner lagrangian point; b) the companion's size is much less than Roche lobe the outflow is connected with a stellar wind. The matter loses part of its kinetic energy in the shock wave and thereafter, gravitational capture of accreting matter becomes possible

The matter that is falling on to the black hole would be having angular momentum which in turn prevents the free fall of matter into the black hole rather the particles would spiral around it and lead to the formation of a disk. The particles in the disk, due to the friction between the adjacent layers, lose their angular momentum and spiral into the black hole, which gets converted into thermal energy. From the spectrum produced, we can determine the rate of accretion.

In this paper they have introduced three major terms:

- 1) **Alpha viscosity parameter:** describes the efficiency and internal friction and turbulence within an accretion disk
- 2) **Mass accretion rate:** Rate at which matter flows into the disk.
- 3) **Eddington luminosity:** it was stated that $L = 10^{38} (M/M_{\odot})$ ergs per seconds was the maximum luminosity that can be attained and the additional mass accreted will be thrown off as gas outflows.

But later Super Eddington Luminosities have been observed disproving this fact

Subcritical Accretion	Supercritical Accretion
Matter flows smoothly and most of it falls into the blackhole.	Here in the region near the blackhole where an effective outflow of matter under the influence of radiation pressure takes place.
Gravity dominates over radiation pressure.	Radiation pressure is dominating here.

From X-rays to Optical Light

The innermost regions of the disk are hot and produce high energy radiation , particularly X-rays and ultraviolet radiation. Some of this radiation travels outward and gets absorbed by the cooler outer regions of the disk, which reprocesses and gets re-emitted in lower energy photons such as Optical. Therefore, optical radiation from a black-hole binary does not necessarily originate directly from the hottest part of the disk. It can be produced through reprocessing of radiation emitted by the inner disk.

The authors point out that the optical spectrum can contain broad recombination and resonance emission lines. They also note that sufficiently strong ultraviolet radiation can ionize the surrounding gas and produce an extended ionized region, or Strömgren region. A Strömgren sphere is defined as the spherical volume of HII regions where the rate of ionizations from hot, young stars balances the rate of recombinations.

A Critical View: What Has Changed Since 1973?

Although the paper is foundational, our understanding of accretion physics has advanced considerably since it was published. Modern models also incorporate effects that were beyond the scope of the original treatment, including general relativity near the black hole, relativistic radiative transfer, Compton scattering, radiation and MHD(magnetohydro dynamics) effects, disk winds and more sophisticated treatments of super-Eddington accretion.